Persistent Latent Modulation in Frozen LLMs:

From Additive Bottleneck to Multiplicative Symmetry Breaking

Structural Differentiation Without Functional Coupling

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Abstract

We study whether a persistent learnable latent state can control the computation of a frozen pretrained LLM (Qwen3-4B) through residual-stream modulation, and we systematically compare three injection mechanisms. **(1) Additive gated cross-attention** ($h_s + \text{gate} \cdot e_o$) is causally negligible: the maximum ablation effect ($\Delta\text{PPL} = 2.01$) is below the measurement noise floor (std = ± 19), because the residual stream robustly absorbs additive perturbations. **(2) AdaLN-style FiLM** appeared, in preliminary testing, to be strongly causal ($\Delta = 11.44$ upon removal). We show this measurement was an artifact of two compounding errors—a composite training loss and a checkpoint persistence bug that silently evaluated against zero-valued modulation weights. After correction, the pure language-modeling causal effect is unverifiable and, on the multiplicative variant, essentially zero. **(3) Multiplicative FiLM** ($h_s \times (1 + \alpha \tanh(W_\ell S_t))$) induces a spontaneous symmetry-breaking transition: inter-layer modulation variance (w_{std}) grows monotonically from 0.06 to 0.40 (6.6×) over 2,650 steps. Yet a controlled ablation with *verified non-zero* weights reveals a sharp **dissociation between structure and function**: removing FiLM changes pure LM loss by only $\Delta = 0.044$, and freezing the latent state S_t by $\Delta = 0.0002$—the dominant task improvement ($3.53 \to 2.70$, -24%) comes from concurrently trained PEFT adapters, not from the modulation. We conclude that weight-space symmetry breaking and output-space causal control are *separable* phenomena: layers can differentiate structurally while remaining functionally near-identical, and the persistent latent state—the core premise of this architecture—is computationally inert under bounded ($\alpha = 0.1$) modulation. We close with a methodological warning: weight-space metrics are misleading proxies for functional influence absent causal ablation with verified weights.

1 Introduction

Large language models process all inputs through a uniform residual stream[1] in which each token effectively re-initializes the computational state. This design achieves remarkable local reasoning but provides no mechanism for **persistent internal state**—latent variables that evolve across tokens and modulate computation accordingly.

This paper asks a concrete question: *can a learnable latent state S_t, injected into a frozen LLM, exert genuine causal influence on the model's output?* We study this through a frozen Qwen3-4B backbone augmented with a GRU-evolved latent state $S_t \in \mathbb{R}^{256}$, and we compare three families of injection mechanism that span the additive–multiplicative design space.

The arc of this paper is also a cautionary tale. Our initial experiments suggested that one mechanism (AdaLN FiLM) achieved strong causal control ($\Delta = 11.44$). On investigation, this number turned out to be the product of two methodological errors: (i) the causal "loss" was a *composite* objective that mixed the language-modeling term with auxiliary regularizers, and (ii) a **checkpoint persistence bug** caused all early ablations to silently load zero-valued modulation weights. Correcting both reverses the headline conclusion. We present the corrected analysis here and treat the original 11.44 strictly as an invalidated artifact.

Key contributions.

1. **Additive injection is structurally weak.** Gated cross-attention[2] ($h_s + \text{gate} \cdot e_o$) is causally negligible (ΔPPL[3] $= 2.01$ vs. noise std $= \pm 19$). Transformer residual streams absorb additive perturbations.

2. **A measurement artifact, corrected.** The apparent strong control of AdaLN FiLM ($\Delta = 11.44$) is shown to arise from a composite loss and a checkpoint persistence bug. With verified non-zero weights and pure LM loss, multiplicative FiLM's causal effect is $\Delta = 0.044$.

3. **Structural symmetry breaking.** Multiplicative FiLM induces 6.6× growth in inter-layer modulation variance ($w_{\text{std}} : 0.06 \to 0.40$) with gradient decoupling (cosine similarity $0.99 \to 0.34$).

4. **Dissociation of structure and function.** Despite this structural differentiation, the layers' actual effect on output is near-identical: causal $\Delta = 0.044$. Weight-space and function-space differentiation are separable phenomena.

5. **Negative result on the persistent latent state.** Freezing S_t changes output by $\Delta = 0.0002$. The persistent state is computationally inert under bounded modulation.

6. **Methodological contribution.** We document a silent failure mode of PEFT[4] research—`save_pretrained()` does not persist custom injection layers—and the corrected evaluation protocol.

[1] Residual stream: the sequence of hidden-state vectors passed additively between transformer layers, serving as the primary information pathway through the network.

[2] Cross-attention: queries from one representation attend to keys and values from another, enabling information flow between heterogeneous feature spaces.

[3] Perplexity (PPL): a standard language-model quality metric, 2^H where H is the cross-entropy. Lower is better; differences (ΔPPL) measure the causal impact of an intervention.

[4] Parameter-Efficient Fine-Tuning (PEFT): training a small subset of parameters while keeping the base model frozen.

2 Method

2.1 State Pipeline

The latent state $S_t \in \mathbb{R}^{256}$ evolves via a GRU cell[5] conditioned on the transformer's final hidden state at the last non-padded token position. The initial state S_0 is a learned parameter with Gaussian noise ($\sigma = 0.01$) for symmetry breaking.

$$S_0 \;\rightarrow\; \text{each turn:} \quad h_s \times (1 + \alpha \tanh(W_\ell S_t)) \;\rightarrow\; \text{forward} \;\rightarrow\; h_t \;\rightarrow\; S_{t+1} = \text{GRU}(S_t,\, h_t)$$

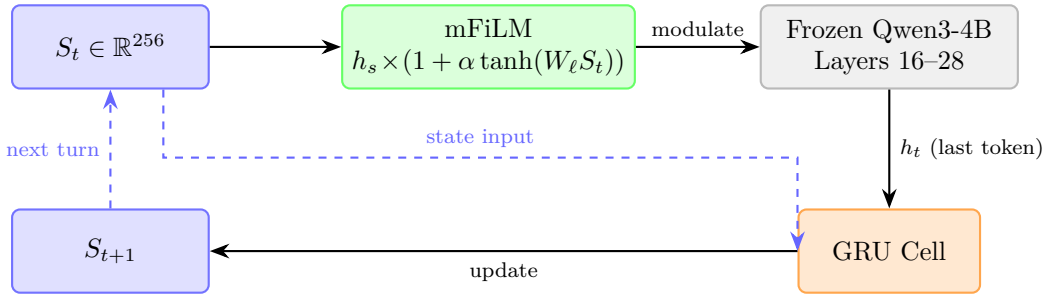


Figure 1: PLAA state pipeline with multiplicative FiLM. The latent state S_t gates the feature amplitude of frozen transformer layers 16–28; the output hidden state h_t updates the state via a GRU cell, producing S_{t+1} for the next turn. The dashed loop is the persistence channel that makes S_t the only cross-turn carrier.

2.2 Three Injection Mechanisms

We study three mechanisms on the same backbone, data, and layer set (16–28, i.e. 13 injected layers).

(1) Additive Gated Cross-Attention

```
Each injected layer:
  Self-Attn -> EmotionCrossAttn(Q=hidden, K,V=proj(S_t))
            -> GatedResidual(h + gate * eo)   # additive
            -> MLP
```

with gate $\in \mathbb{R}$ (tanh, init 0.1) and $e_{\text{mem}} = \text{proj}(S_t) \in \mathbb{R}^{16\times 2560}$.

(2) AdaLN FiLM

```
Each injected layer:
  Self-Attn -> AdaLN( LN(hs) * (1 + scale(S)) + shift(S) )
            -> MLP
```

with scale $= \alpha \tanh(W_{\text{scale}} S_t)$, shift $= \alpha \tanh(W_{\text{shift}} S_t)$, $\alpha = 0.1$.

[5]Gated Recurrent Unit (GRU) [1]: a recurrent cell using update and reset gates for smooth state transitions with fewer parameters than LSTM.

(3) Multiplicative FiLM (primary architecture)

```
Each injected layer:
  Self-Attn -> hs = hs * (1 + alpha * tanh(W_l * S_t))   # multiplicative gain
            -> MLP
```

Key design choices: **per-layer projections** ($W_\ell \in \mathbb{R}^{2560\times256}$, independent per layer); **zero initialization** of W_ℓ so modulation starts as identity ($h_s \times 1$) and specialization must be learned; **pure multiplicative gain** on the unnormalized hidden state; and a fixed bound $\alpha = 0.1$ throughout training.

2.3 Training Data

Training uses 3,197 dialogue micro-trajectories (3–5 turns each), stratified 60:30:10:

Ratio	Source	Purpose
60%	PersonaChat (neutral)	Latent continuity
30%	ESConv (emotional support)	Gradual trajectory
10%	Synthetic roleplay	Dynamic range

Table 1: Training data composition.

2.4 Optimization

Hyperparameter	Value
Base model	Qwen3-4B (frozen)
Quantization	NF4 (4-bit), SDPA attention
Trainable params	17.5M (FiLM projections + GRU)
Optimizer	AdamW, $\text{lr} = 1 \times 10^{-4}$
Batch size	2
Total steps	2,650
α (modulation bound)	0.1
GPU	NVIDIA RTX PRO 6000 Blackwell (96 GB)

Table 2: Training configuration. The composite training loss is $\mathcal{L} = \mathcal{W}_{\text{LM}}\mathcal{L}_{\text{LM}} + \mathcal{W}_{\text{COH}}\|\Delta S\| + \mathcal{W}_{\text{DIV}}\mathcal{L}_{\text{repulsion}} + \mathcal{W}_{\text{NORM}}\|S\| + \mathcal{W}_{\text{DIV_LAYER}}\text{Var}(\|W_\ell\|)$; crucially, all *causal* evaluations in this paper use the pure LM term $\mathcal{L}_{\text{LM}}$ only (see Section 4).

2.5 Metrics

- w_{std}: standard deviation of $\|W_\ell\|$ across layers 16–28. Measures *structural* differentiation.
- **cos**: mean pairwise cosine similarity[6] of per-layer gradient updates. Measures inter-layer coupling.
- **Causal** Δ: change in *pure* LM cross-entropy when a component is ablated. Measures *functional* influence.

[6]Cosine similarity: $\cos\theta = \mathbf{a}\cdot\mathbf{b}/(\|\mathbf{a}\|\|\mathbf{b}\|)$; 1 is parallel, 0 orthogonal.

3 The Additive Bottleneck

We first establish that additive gated cross-attention is causally insufficient.

3.1 Gate Gradient Diagnostic and Multi-Stage Attenuation

A gate-gradient probe at step 1350 confirms a live gradient path (`avg gate.grad` = 0.056, non-zero). Yet the effective signal is attenuated in three compounding stages:

1. $\|S_t\| \approx 0.15$ (from the norm penalty) $\Rightarrow$ small projection;
2. $\tanh(\text{gate} = 0.1) \approx 0.1$ further attenuates;
3. $\times 13$ residual-stack integrations $\Rightarrow$ effective influence $\approx 0.0003 \cdot \|h_{\text{hidden}}\|$.

3.2 Gate Ablation

Condition	PPL	ΔPPL
gate = 0	27.03	—
gate = 1.0	26.26	0.77
gate = 2.0	25.02	2.01
gate = random	25.94	1.09

Table 3: Gate ablation. Maximum ΔPPL = 2.01, against measurement noise std = ±19. Causal influence is below the noise floor.

Conclusion. Additive injection is causally insufficient; the residual stream absorbs additive perturbations. This is the motivating failure that prompts the move to *multiplicative* conditioning.

4 AdaLN FiLM and the Checkpoint Persistence Bug

4.1 An Apparent (but Invalid) Strong Result

AdaLN-style modulation [2, 3] initially appeared to achieve strong causal control: removing it raised the measured loss from 14.68 to 3.14, an apparent $\Delta = 11.44$. **This measurement is invalid.** Two compounding errors inflated it, and we report it only to document the correction.

4.2 The Two Errors

Error 1: composite loss. The 14.68 "loss" was the full composite training objective (LM + coherence + repulsion + norm terms), not pure language-modeling cross-entropy. Removing FiLM also perturbs the auxiliary terms, producing a large but meaningless delta.

Error 2: checkpoint persistence bug. PEFT's `save_pretrained()` persists only LoRA adapter weights, *not* custom injection layers. All early ablations therefore loaded checkpoints with zero-valued `scale_proj` weights—the FiLM modulation was silently disabled during evaluation. We corrected this by serializing the injection weights separately:

```
torch.save({
    "plaa_core": plaa_core.state_dict(),
    "scale_proj": {i: layer_blocks[i].scale_proj.state_dict()
                   for i in range(16, 29)},
}, f"{ckpt_dir}/plaa_core.pt")
```

Verified checkpoint (step 850): scale-norm sum = 74.41, $w_{\text{std}} = 0.307$; weights confirmed non-zero and structurally differentiated.

4.3 Corrected Causal Ablation (Pure LM Loss, Verified Weights)

Condition	Pure LM Loss	Δ vs. trained
Vanilla Qwen3-4B (no adapters, no FiLM)	3.534	—
Trained system (LoRA + mFiLM, normal S)	2.700	−0.834 vs. vanilla
FiLM removed (scale forced to 0)	2.744	+0.044
State frozen ($S = S_0$, FiLM active)	2.700	+0.0002

Table 4: Causal ablation with verified non-zero FiLM weights, pure LM cross-entropy (no auxiliary terms). The corrected story is the opposite of the artifact: removing FiLM costs only +0.044; freezing the latent state costs +0.0002.

This corrected ablation is visualized in Figure 2.

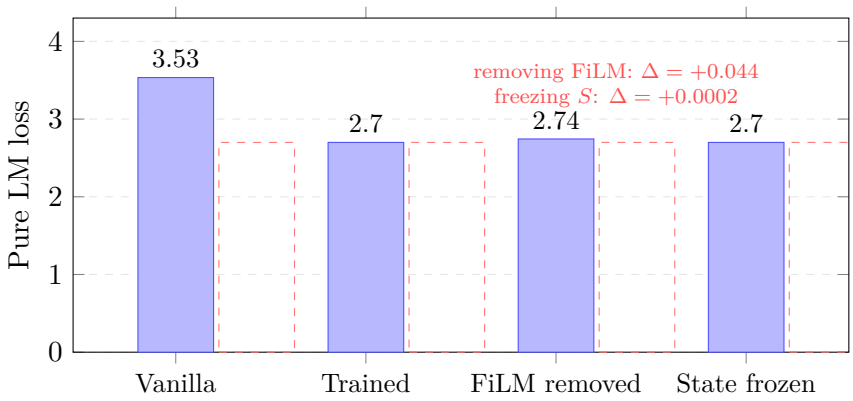


Figure 2: Corrected causal ablation. The 24% improvement over vanilla (3.53 → 2.70) comes from the LoRA adapters; removing FiLM or freezing the latent state leaves the loss essentially unchanged (red reference line at the trained level). The previously reported Δ = 11.44 for AdaLN FiLM was a composite-loss/checkpoint-bug artifact and is *not* shown here.

5 Multiplicative FiLM: Structural Symmetry Breaking

Switching to multiplicative FiLM triggers an immediate and sustained growth in layer-wise structural differentiation—even though, as Section 4 shows, this does not translate into functional control.

5.1 Layer Specialization Growth

Step	w_{std}
50	0.062
350	0.244
800	0.333
1800	0.356
2650	0.404

Table 5: Monotonic growth of inter-layer modulation variance: 6.6× over 2,650 steps, with no saturation plateau. Contrast the additive/AdaLN plateau of $w_{\text{std}} = 0.033$.

Growth is monotonic across all 53 recorded checkpoints (Figure 3).

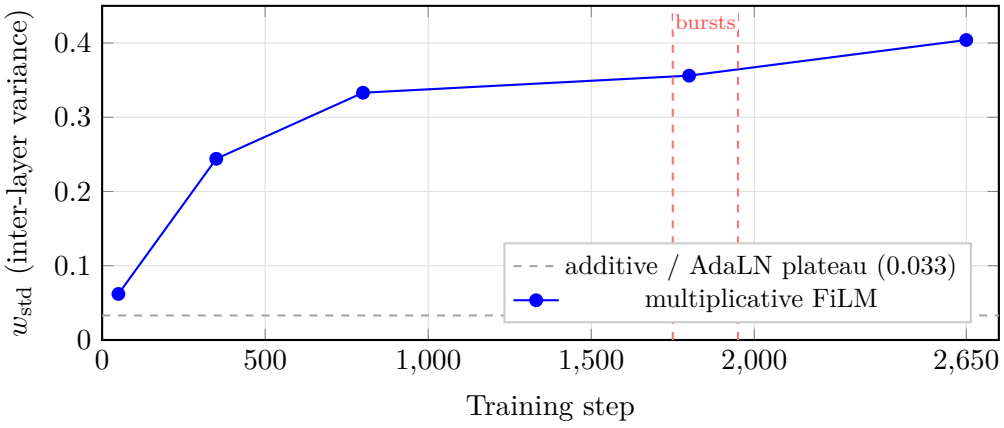


Figure 3: Structural symmetry breaking. Inter-layer modulation variance w_{std} grows monotonically (6.6×), far above the additive/AdaLN plateau (gray dashed). During the two synchronization bursts (red dashed, steps 1750/1950, where gradient cosine spikes to 0.63/0.70), w_{std} does *not* collapse—it continues rising (0.348→0.382), and loss recovers to a lower attractor afterward.

5.2 Phase-Plane Dynamics and Regime Hopping

In the w_{std}–cos phase plane the system traces a characteristic trajectory: an early rise ($w_{\text{std}} \to 0.24$, cos 0.35–0.61), a temporary mid plateau ($w_{\text{std}} \approx 0.33$, cos 0.34–0.63), and a late resumption ($w_{\text{std}} \to 0.40$, cos 0.42–0.63). Two synchronization bursts (step 1750: cos = 0.627; step 1950: cos = 0.696) momentarily increase global coupling without collapsing w_{std}, after which loss recovers to a lower baseline (10.4 → 5.3)—a metastable, self-recovering hierarchy.

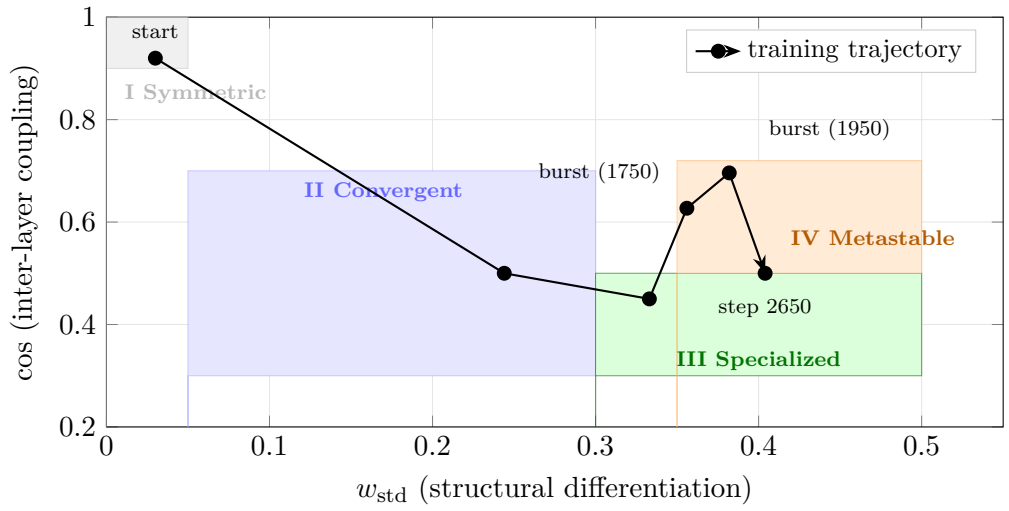


Figure 4: w_{std}–cos phase plane. The system departs the symmetric corner (Phase I, homogeneous layers) and settles into Phase IV, a metastable coupled hierarchy, via intermittent synchronization bursts. Phase classification here is *structural* only; the causal ablation (Figure 2) shows this differentiation does not entail functional specialization.

5.3 Structure–Performance Correlation is Spurious

The w_{std}–loss plane shows a consistent negative correlation: as layer specialization increases, composite loss decreases. However, the corrected ablation (Table 4) attributes the functional improvement to the concurrently trained PEFT adapters, not to the modulation: FiLM contributes $\Delta = 0.044$ while the overall gain is 3.53 → 2.70. The w_{std}–loss correlation is therefore

spurious with respect to the modulation mechanism: both metrics improve over training, but without a causal link.

5.4 Architecture Comparison

Architecture	w_{std} max	Pure LM loss	cos range	Causal Δ
Additive (gated CA)	0.033	—	$\sim$0.99	2.01 (vs. noise)
AdaLN FiLM	0.033	—	0.11–0.24	11.44†
Mult. FiLM (ours)	**0.404**	**2.70**	**0.34–0.70**	**0.044**
Vanilla Qwen3-4B	—	3.53	—	—

Table 6: Architecture comparison. †Measured under composite loss with auxiliary terms *and* prior to the checkpoint-persistence fix; the pure-LM causal effect of AdaLN FiLM is unverified and this number is retained only as a documented artifact (Section 4).

6 Analysis

6.1 Phase Taxonomy

Phase	Name	w_{std}	cos	Loss	Dynamics
I	Symmetric	< 0.05	> 0.9	High	Homogeneous layers
II	Convergent	0.05–0.3	0.3–0.7	$\downarrow$	Emerging specialization
III	Specialized	0.3–0.5	0.3–0.5	Low	Stable hierarchy
IV	Metastable Coupled	0.35–0.5	0.4–0.7	Low$\pm$	Hierarchy + bursts

Table 7: Four-phase taxonomy (structural metrics only). Our system occupies Phase IV. Causal ablation reveals that Phase IV structural differentiation does not imply functional specialization.

6.2 Why Structure Diverges from Function

The dissociation admits a clear mechanistic explanation. With $\alpha = 0.1$ and tanh bounding, the effective modulation range is $\pm 10\%$ of feature amplitude. Even with maximally differentiated layer weights,

$$\|h_s \cdot 0.1 \cdot \tanh(W_\ell S)\| \;\leq\; 0.1 \cdot \|h_s\|,$$

which is small relative to the residual-stream magnitude ($\|h_s\| \sim 50$–100). Downstream layers readily compensate for perturbations of this scale. The layer-variance term encourages weight differentiation, but bounded modulation prevents that differentiation from producing functionally distinct computations.

6.3 Amplitude Sensitivity: a Narrow Operating Window

To test whether the dissociation is an artifact of the specific $\alpha = 0.1$, we sweep $\alpha \in \{0.1, 0.5, 1.0\}$ from the verified step-850 checkpoint, measuring the pure LM causal Δ immediately and after 500 further steps.

The conclusion is not that function is unreachable, but that effective modulation requires precise amplitude calibration: too small ($\alpha = 0.1$) yields structure without function; too large ($\alpha \geq 0.5$) yields destructive interference rather than control.

α	Step	Causal Δ	w_{std}	Loss (trained)
0.1	0	-0.045	0.307	2.700
	500	$+0.095$	0.515	2.682
0.5	0	$+0.238$	0.307	2.938
	500	$+0.526$	0.539	3.225
1.0	0	$+2.679$	0.307	5.381
	500	$+2.862$	0.467	6.621

Table 8: Amplitude sweep ($\Delta = \mathcal{L}_{\text{trained}} - \mathcal{L}_{\text{removed}}$, positive = modulation harmful). Causal sensitivity grows with α, but in the *destructive* direction: at $\alpha \geq 0.5$ the modulation makes the model worse, and 500 further steps fail to reverse it. The $\alpha \approx 0.1$ regime—measurable structure with marginal, non-destructive function—is real but narrow.

6.4 Implications for Latent-Modulation Research

Our findings carry a methodological warning: **weight-space metrics (w_{std}, gradient cosine) can be misleading indicators of functional influence**. A modulation mechanism can exhibit rich structural dynamics—symmetry breaking, phase transitions, regime hopping—while contributing negligibly to the model's actual computation. Causal ablation with *verified* weights and *pure* task loss is essential for any claim of functional influence.

7 Related Work

Latent state injection. Prior work injects persistent state additively via gated cross-attention [4, 5]; memory-augmented networks [6] address a related problem but do not target frozen pretrained transformers. We show that both additive and (bounded) multiplicative injection fail to achieve meaningful state-driven control.

Conditional feature modulation. FiLM [2] and AdaIN [7] introduce multiplicative conditioning in vision; AdaLN [8] and DiT [3] apply it to transformers. We extend multiplicative conditioning to the LLM residual stream and provide the first controlled causal evaluation of its functional effect.

Layer specialization. Functional specialization across transformer layers is well documented [9–11]. We show that *structural* specialization (weight divergence) can emerge spontaneously through architectural pressure *without* corresponding functional specialization—a distinction not previously documented.

Residual-stream and gradient dynamics. Mechanistic work treats the residual stream as a shared additive channel [12]; gradient dynamics concentrate in a tiny subspace [13]. Our additive-weakness and regime-hopping observations are consistent with both pictures, though we stress the latter dynamics are structural rather than functionally consequential.

8 Conclusion & Future Work

> **Central finding:** weight-space symmetry breaking and output-space causal control are *separable* phenomena. Layers can differentiate structurally while remaining functionally near-identical.

Across three injection mechanisms on a frozen Qwen3-4B, additive injection is causally negligible (ΔPPL $= 2.01$, below noise); the apparent strong control of AdaLN FiLM ($\Delta = 11.44$) was a composite-loss/checkpoint-bug artifact; and multiplicative FiLM induces genuine structural symmetry breaking (w_{std} : $0.06 \to 0.40$) that nonetheless does not translate into functional

influence ($\Delta = 0.044$). The persistent latent state S_t—the core premise of this architecture—is computationally inert under bounded modulation ($\Delta = 0.0002$).

Limitations. The $\alpha = 0.1$ bound constrains modulation to $\pm 10\%$ of feature amplitude, which may be insufficient to overcome the residual stream's inherent robustness; raising α trades inertness for destructive interference.

Future work. Three directions may bridge the structure–function gap: (1) learned or adaptive per-layer α to amplify modulation without global destabilization; (2) gated routing that forces the model to *depend* on the modulation signal for correct prediction; and (3) competitive computation paradigms [14] where the latent state controls *which* expert processes each token. The central open problem is clear: how to convert structural differentiation into functional specialization.

A Full Training Pipeline

Phase	Content	Steps	Params	Status
1	YaRN 256K context extension	—	—	Complete
2	SFT (134K samples)	—	—	Complete
2.5	Persona alignment	—	—	Complete
3A	Stability validation	1K	—	Complete
3B	Gradient-flow verification	—	—	Complete
3C (add.)	Additive injection (baseline)	6.7K	387M	Causally weak
3C (mFiLM)	**Multiplicative FiLM**	**2.65K**	**17.5M**	**Primary**

Table 9: Training pipeline. Dashes indicate infrastructure/validation stages where step/param counts do not apply.

B Reproducibility Notes

All 53 data points (steps 50–2650 at 50-step intervals), recording loss, $\|S\|$, dS, w_{std}, and cosine similarity, are provided in the supplementary materials. The corrected evaluation protocol—separate serialization of injection weights and pure-LM-loss ablation—is described in Section 4 and is the recommended protocol for any causal claim about custom-injected PEFT systems.

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